

# HAT-P-32b

R-0263

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-32\_250830 · created 2026-07-17T09:23:43+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460917.7799 +/- 0.0047 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.186 +/- 0.052                 |
| Transit depth (Rp/Rs)^2                           | 3.45 +/- 1.94 [%]               |
| Orbital Inclination (inc)                         | 84.47 +/- 1.75                  |
| Airmass coefficient 1 (a1)                        | 1.121 +/- 0.062                 |
| Airmass coefficient 2 (a2)                        | -0.077 +/- 0.036                |
| Scatter in the residuals of the lightcurve fit is | 5.7 %                           |
| Best Comparison Star                              | #3 - [413, 291]                 |
| Optimal Aperture                                  | 2.49                            |
| Optimal Annulus                                   | 13.46                           |
| Transit Duration (day)                            | 0.1134 +/- 0.007                |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.186 ± 0.052 vs 0.1489 (deviation 0.71σ) |
| Duration fit vs published | 0.1134 d vs 0.1304 d (deviation 2.43σ)    |
| Mid-time O–C              | 11.9 min                                  |
| Depth SNR                 | 3.6                                       |
| Residual scatter          | 5.7 %                                     |

## Light curve

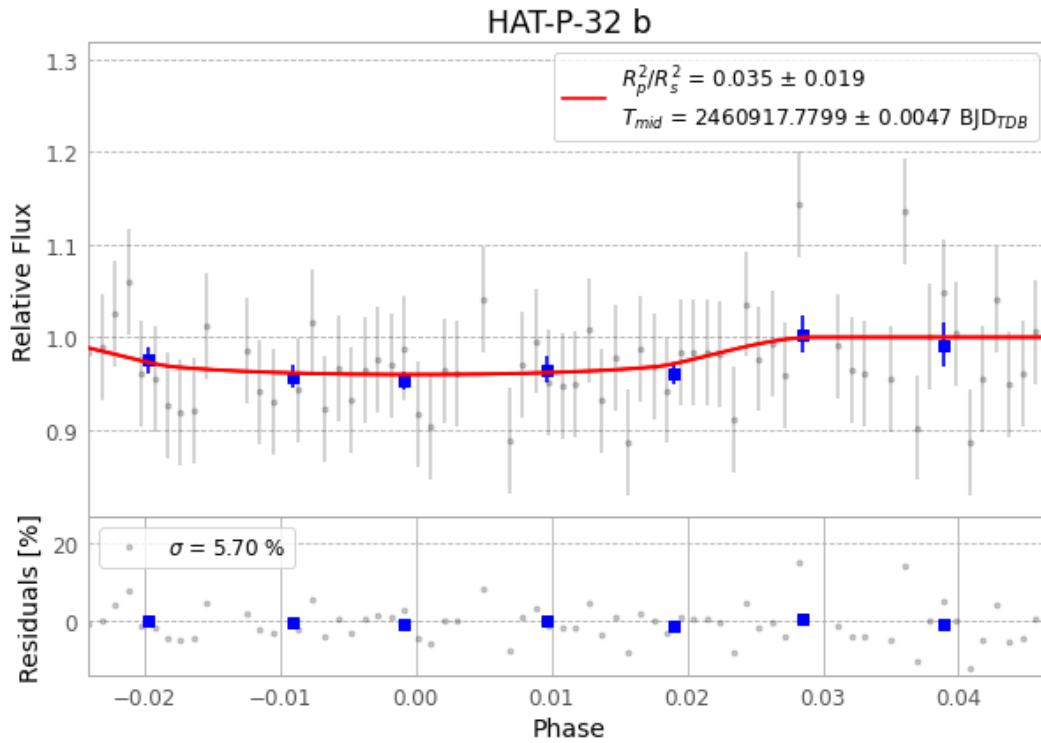


Figure 1 — FinalLightCurve\_HAT-P-32 b\_2025-08-29.png (SHA-256 8f848a68af2b3d35...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 155], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 e0b515d1dcd033a9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 7dbc7ef

## Data provenance

74 FITS frames (49 MB), HATP-32250830052416.FITS → HATP-32250830090315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-250830.log (34ebb0ff2bea...), FinalLightCurve\_HAT-P-32 b\_2025-08-29.png (8f848a68af2b...), FinalLightCurve\_HAT-P-32 b\_2025-08-29.csv (fdd82b5e3ae6...)

## Compute resources

Wall time 215.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 09:23Z.